

ilarly. Obviously, we always have $rvc(G) \leq n - 2$ (except for the singleton graph), and $rvc(G) = 0$ if and only if G is a clique. Also clearly, $rvc(G) \geq diam(G) - 1$ with equality if the diameter of G is 1 or 2.

Note that $rvc(G)$ may be much smaller than $rc(G)$ for some graph G . For example, $rvc(K_{1,n-1}) = 1$ while $rc(K_{1,n-1}) = n - 1$. $rvc(G)$ may also be much larger than $rc(G)$ for some graph G . For example, take n vertex-disjoint triangles and, by designating a vertex from each of them, add a complete graph on the designated vertices. This graph has n cut-vertices and hence $rvc(G) \geq n$. In fact, $rvc(G) = n$ by coloring only the cut-vertices with distinct colors. On the other hand, it is not difficult to see that $rc(G) \leq 4$. Just color the edges of the K_n with, say, color 1, and color the edges of each triangle with the colors 2, 3, 4.

In Section 2, we will focus on the rainbow connection number and strong rainbow connection number. We collect many upper bounds for these two parameters. From Section 3 to Section 5, we survey on the other three parameters: rainbow k -connectivity, k -rainbow index, rainbow vertex-connection number, respectively. In the last section, we sum up the results on algorithms and computational complexity.

1.2 Terminology and notations

All graphs considered in this survey are finite, simple and undirected. We follow the notations and terminology of [7] for all those not defined here. We use $V(G)$ and $E(G)$ to denote the set of vertices and the set of edges of G , respectively. For any subset X of $V(G)$, let $G[X]$ denote the subgraph induced by X , and $E[X]$ the edge set of $G[X]$; similarly, for any subset F of $E(G)$, let $G[F]$ denote the subgraph induced by F . Let $\mathcal{G}$ be a set of graphs, then $V(\mathcal{G}) = \bigcup_{G \in \mathcal{G}} V(G)$, $E(\mathcal{G}) = \bigcup_{G \in \mathcal{G}} E(G)$. We define a *clique* in a graph G to be a complete subgraph of G , and a *maximal clique* is a clique that is not contained in any larger clique of G . For a set S , $|S|$ denotes the cardinality of S . An edge in a connected graph is called a *bridge*, if its removal disconnects the graph. A graph with no bridges is called a *bridgeless* graph. A vertex is called *pendant* if its degree is 1. We call a path of G with length k a *pendant k -length path* if one of its end vertex has degree 1 and all inner vertices have degree 2 in G . By definition, a pendant k -length path contains a pendant ℓ -length path ($1 \leq \ell \leq k$). A pendant 1-length path is a *pendant edge*. We denote C_n a cycle with n vertices. For $n \geq 3$, the *wheel* W_n is constructed by joining a new vertex to every vertex of C_n . We use $g(G)$ to denote the girth of G , that is, the length of a shortest cycle of G .

Let G be a connected graph. Recall that the distance between two vertices u and v in G , denoted by $d(u, v)$, is the length of a shortest path between them in G . The *eccentricity* of a vertex v is $ecc(v) := \max_{x \in V(G)} d(v, x)$. The *diameter* of G is $diam(G) := \max_{x \in V(G)} ecc(x)$. The *radius* of G is $rad(G) := \min_{x \in V(G)} ecc(x)$. Distance between a vertex v and a set $S \subseteq V(G)$ is $d(v, S) := \min_{x \in S} d(v, x)$. The *k -step open neighbourhood* of a set $S \subseteq V(G)$ is $N^k(S) := \{x \in V(G) | d(x, S) = k\}$, $k \in \{0, 1, 2, \dots\}$. A set $D \subseteq V(G)$ is called a *k -step dominating set* of G , if every vertex in G is at a distance at most k from D . Further, if D

induces a connected subgraph of G , it is called a *connected k -step dominating set* of G . The cardinality of a minimum connected k -step dominating set in G is called its *connected k -step domination number*, denoted by $\gamma_c^k(G)$. We call a two-step dominating set *k -strong* [36] if every vertex that is not dominated by it has at least k neighbors that are dominated by it. In [11], Chandran, Das, Rajendraprasad and Varma made two new definitions which will be useful in the sequel. A dominating set D in a graph G is called a *two-way dominating set* if every pendant vertex of G is included in D . In addition, if $G[D]$ is connected, we call D a *connected two-way dominating set*. A (connected) two-step dominating set D of vertices in a graph G is called a (connected) *two-way two-step dominating set* if (i) every pendant vertex of G is included in D and (ii) every vertex in $N^2(D)$ has at least two neighbours in $N^1(D)$. Note that if $\delta(G) \geq 2$, then every (connected) dominating set in G is a (connected) two-way dominating set.

A subgraph H of a graph G is called *isometric* if distance between any pair of vertices in H is the same as their distance in G . The size of a largest isometric cycle in G is denoted by $iso(G)$. A graph is called *chordal* if it contains no induced cycles of length greater than 3. The *chordality* of a graph G is the length of a largest induced cycle in G . Note that every isometric cycle is induced and hence $iso(G)$ is at most the chordality of G . For $k \leq \alpha(G)$, we use $\sigma_k(G)$ to denote the minimum degree sum that is taken over all independent sets of k vertices of G , where $\alpha(G)$ is the number of elements of a maximum independent set of G .

2 (Strong) Rainbow connection number

2.1 Basic results

In [12], Chartrand, Johns, McKeon and Zhang did some basic research on the (strong) rainbow connection numbers of graphs. They determined the precise (strong) rainbow connection numbers of several special graph classes including trees, complete graphs, cycles, wheel graphs, complete bipartite graphs and complete multipartite graphs.

Proposition 2.1 [12] *Let G be a nontrivial connected graph of size m . Then*

- (a) $rc(G) = 1$ if and only if G is complete, $src(G) = 1$ if and only if G is complete;
- (b) $rc(G) = 2$ if and only if $src(G) = 2$;
- (c) $rc(G) = m$ if and only if G is a tree, $src(G) = m$ if and only if G is a tree.

Proposition 2.2 [12] *For each integer $n \geq 4$, $rc(C_n) = src(C_n) = \lceil \frac{n}{2} \rceil$.*

Proposition 2.3 [12] *For each integer $n \geq 3$, we have*

$$rc(W_n) = \begin{cases} 1 & \text{if } n = 3, \\ 2 & \text{if } 4 \leq n \leq 6, \\ 3 & \text{if } n \geq 7. \end{cases}$$

and $src(W_n) = \lceil \frac{n}{3} \rceil$.

Proposition 2.4 [12] For integers s and t with $2 \leq s \leq t$, $rc(K_{s,t}) = \min\{\lceil \sqrt[s]{t} \rceil, 4\}$, and for integers s and t with $1 \leq s \leq t$, $src(K_{s,t}) = \lceil \sqrt[s]{t} \rceil$.

Proposition 2.5 [12] Let $G = K_{n_1, n_2, \dots, n_k}$ be a complete k -partite graph, where $k \geq 3$ and $n_1 \leq n_2 \leq \dots \leq n_k$ such that $s = \sum_{i=1}^{k-1} n_i$ and $t = n_k$. Then

$$rc(G) = \begin{cases} 1 & \text{if } n_k = 1, \\ 2 & \text{if } n_k \geq 2 \text{ and } s > t, \\ \min\{\lceil \sqrt[s]{t} \rceil, 3\} & \text{if } s \leq t. \end{cases}$$

and

$$src(G) = \begin{cases} 1 & \text{if } n_k = 1, \\ 2 & \text{if } n_k \geq 2 \text{ and } s > t, \\ \lceil \sqrt[s]{t} \rceil & \text{if } s \leq t. \end{cases}$$

By Proposition 2.1, it follows that for every positive integer a and for every tree T of size a , $rc(T) = src(T) = a$. Furthermore, for $a \in \{1, 2\}$, $rc(G) = a$ if and only if $src(G) = a$. If $a = 3, b \geq 4$, then by Proposition 2.3, $rc(W_{3b}) = 3$ and $src(W_{3b}) = b$. For $a \geq 4$, we have the following.

Theorem 2.6 [12] Let a and b be positive integers with $a \geq 4$ and $b \geq \frac{5a-6}{3}$. Then there exists a connected graph G such that $rc(G) = a$ and $src(G) = b$.

Then, combining Propositions 2.1 and 2.3 with Theorem 2.6, they got the following result.

Corollary 2.7 [12] Let a and b be positive integers. If $a = b$ or $3 \leq a < b$ and $b \geq \frac{5a-6}{3}$, then there exists a connected graph G such that $rc(G) = a$ and $src(G) = b$.

Finally, they thought the question that whether the condition $b \geq \frac{5a-6}{3}$ can be deleted? and raised the following conjecture:

Conjecture 2.8 [12] Let a and b be positive integers. Then there exists a connected graph G such that $rc(G) = a$ and $src(G) = b$ if and only if $a = b \in \{1, 2\}$ or $3 \leq a \leq b$.

In [19], Chen and Li gave a confirmative solution to this conjecture by showing a class of graphs with given rainbow connection number a and strong rainbow connection number b .

From the above several propositions, we know $rc(G) = src(G)$ hold for some special graph classes. A difficult problem is following: